Impact of Crowding on Diffusive Search: A Fokker–Planck and First Passage Time Approach

Nirab Hossain, Zachary Kilpatrick Summer 2026

Motivation. A central challenge in biological search is understanding how crowding shapes foraging efficiency. A honeybee navigating within a nest encounters spatially varying traffic density, which reduces its effective diffusivity in congested regions through repeated interactions with nearby individuals. Such crowding-mediated transport arises in trophallaxis networks [Gharooni Fard et al., 2020], where local density governs how efficiently resources spread through the colony, and in queen-finding behavior [Nguyen et al., 2021], where bees must locate a target despite spatial heterogeneity in mobility and signaling. Crucially, crowding can also enhance target detection: slower motion near a resource extends interaction time, potentially improving recognition. This creates a trade-off between transport and detection. Reduced diffusivity delays arrival but may improve discoverability. This raises a nontrivial question: *are there regimes in which crowding facilitates search?* This question is directly motivated by high-resolution video data of honeybees foraging in an artificial hive, collected by our collaborator Prof. Orit Peleg (BioFrontiers Institute, CU Boulder).

Model Overview and Goals. We study a diffusive search process with spatially varying diffusivity $D(x)$ on a periodic domain containing a partially absorbing target of strength κ. Using the backward Fokker–Planck framework, the mean first passage time (MFPT) decomposes into a dwell time at the target and a travel time determined by $D(x)$. With constant κ these contributions are decoupled and minimizing MFPT reduces to optimizing transport alone. The problem becomes richer when target discoverability depends on local diffusivity, e.g. $\kappa = \kappa_0/D(0)$, motivated by increased recognition time at low mobility. The central scientific goal is to identify regimes in which crowding *facilitates* target search. Concretely, the project will: (i) derive and analyze the MFPT for spatially varying diffusivity and characterize the transport–detection trade-off; (ii) compute explicit results for representative profiles (cosine, piecewise-linear) and determine whether an optimal diffusivity profile exists; (iii) extend to a *dynamically evolving* diffusivity, in which the Fokker–Planck equation governs a searcher population whose progressive depletion as individuals find the target feeds back on $D(x,t)$; and (iv) validate results numerically and compare with Prof. Peleg's artificial-hive data.

Novel Mathematical Contributions. The primary novelty lies in the treatment of the coupled, time-evolving system in goal (iii). When $D(x,t)$ evolves slowly relative to individual search trajectories, with profiles such as an inverted Gaussian centered at the trap, we may exploit a separation of timescales via an adiabatic asymptotic approximation. The flux into the target is computed with D frozen at each instant, and the slow evolution of D is tracked self-consistently through its dependence on the surviving population fraction. This yields a tractable reduction of an otherwise nonlinear Fokker–Planck system and generates explicit, parameterizable predictions for how search efficiency evolves as the hive empties.

Expected Outcomes. The project will produce (i) explicit MFPT formulas under static heterogeneous diffusivity, characterizing crowding-facilitated search regimes; (ii) an asymptotic theory for the dynamically coupled case, yielding experimentally testable predictions; and (iii) numerical validation of both components. Results will be submitted to *Journal of Physics A* or *New Journal of Physics* and will contribute to Nirab's dissertation. Predictions will be tested against analyzed high-resolution video data collected from flexible artificial hive experiments carried out in Prof. Peleg's lab.

Feasibility. This project builds on substantial existing momentum. Nirab developed the Fokker–Planck and MFPT framework for spatially varying diffusivity as part of a preliminary project in Prof. Rodriguez's APPM 6470 (Spring 2026), including derivation of the backward equation and explicit MFPT formulas, and has working notes on the static problem. The extensions such as competitive detection, adiabatic approximation for evolving $D(x,t)$ and experimental comparison are well-scoped for a focused summer effort.

Requested Support: I request 1 month of summer RA support at 50% effort.

References

Gharooni Fard, G., Bradley, E., & Peleg, O. (2020). Data-driven modeling of resource distribution in honeybee swarms. *Proc. ALIFE 2020*, 324–332.

Nguyen, D. M. T., Iuzzolino, M. L., Mankel, A., Bozek, K., Stephens, G. J., & Peleg, O. (2021). Flow-mediated olfactory communication in honeybee swarms. *Proc. Natl. Acad. Sci. USA*, 118(13), e2011916118.

Endorsement by Z.P. Kilpatrick. Nirab is a highly motivated and hardworking first-year PhD student. Since the beginning of the academic year he has been meeting with me once or twice a month and engaging thoughtfully with background literature related to this project. The primary reason he has not pursued it more intensively to this point is that I have deliberately encouraged him to prioritize his preliminary examinations. He passed his PDE preliminary exam in January and has a solid plan for preparing for the Numerical Analysis exam in May. If he follows that plan, I am confident he will complete all prelims by May and be ready to begin summer research in earnest in late May or early June.

This project is an excellent fit for Nirab's dissertation trajectory. It sits squarely at the interface of PDE theory, asymptotic methods, and mathematical biology, and the work he completed in APPM 6470 gives him genuine momentum. The adiabatic asymptotic framework we will develop here will form an extendable and reusable analytical toolkit for subsequent dissertation chapters. We anticipate the results will be suitable for *Journal of Physics A* or *New Journal of Physics*, and that Nirab will present this work at venues such as Dynamics Days and the SIAM Conference on Dynamical Systems in 2027.

Our collaboration with Prof. Peleg's group is a key part of this project. She brings a rare combination of experimental precision and deep physical and mathematical intuition, having trained as a soft matter physicist, and her group's artificial-hive dataset is unique. Over the summer, our broader group will also host two high school students engaged in outreach activities, including fabricating a board game on collective behavior for family science nights at local high schools in the fall. Nirab's enthusiasm and interpersonal skills make him a natural mentor for these students, though I view this as secondary to ensuring he gets a strong start on research.

I am fully committed to mentoring Nirab through both the analytical and computational components of this project and plan to maintain close contact over the summer with iterative feedback on writing and analysis. I hold a small NSF grant directly relevant to this work; however, the award was reduced considerably at the time of funding and does not stretch far enough to provide substantial graduate student support, covering primarily some travel and a portion of my own summer salary. The department summer RA mechanism is therefore an important and genuinely needed supplement, and would provide Nirab with dedicated time to build the momentum this project deserves at the outset of his dissertation.